c. Breathing roots or Respiratory roots

Avicennia is a tree which grows in mangroves or swamps. They have roots which are seen above the ground for the purpose of gaseous exchange. These roots are erect, peg like structures with numerous pores through which air circulates. These roots are called **breathing roots or pneumatophores.**



Avicennia



Vanda is an epiphytic plant, which grows on trees. The velamen tissue present in the epiphytic root absorbs moisture to perform photosynthesis.



Vanda

d. Haustorial roots

Roots may also perform some special functions. Haustoria or **sucking roots**, are one such example. Cuscuta a parasite plant, climb the trees and other vegetation and use the haustorial roots to penetrate the tissue of the host plant and suck nutrients from them. They are usually found in parasitic plants that depend on the host plants for nutrients.



Cuscuta

5.4.2 Modification of stems

Can you guess what is common between ginger, onion bulb and potatoes. All three are stems. Some plants have their stems modified for storing food and for vegetative propagation. Modified stem may be aerial, subaerial or underground stems.

a. Aerial Modifications

Phylloclade

In dry climate, conserving water is a challenge. Water evaporates from the surface. If the surface area is larger, evaporation would be more and if the surface area is smaller, the evaporation will be less. Plants with many leaves have more surface area. **Cactus** hence



Cactus

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